

Chapter 19 – Cloud Computing

Introduction to Cloud Computing

◆ What is Cloud Computing?

Cloud Computing means:

 Accessing IT resources, applications, servers, storage, and services through the internet from anywhere in the world.

Instead of keeping everything on your own physical computer or office server, you use resources hosted on remote cloud servers.

Simple Definition

Cloud computing allows users and companies to:

- ✓ Use servers
- ✓ Store data
- ✓ Run applications
- ✓ Host websites
- ✓ Create virtual machines
- ✓ Use databases

without buying and managing physical hardware.

Real-Life Example

Examples of cloud services you use daily:

Service	Cloud Example
 Email	Gmail, Outlook
 File Storage	Google Drive, OneDrive
 Streaming	Netflix

Service	Cloud Example
 Meetings	Microsoft Teams, Zoom
 Photo Backup	iCloud, Google Photos

All these services work using cloud infrastructure.

Main Concept of Cloud

“You can access resources from anywhere.”

Meaning:

- You can log in from any device
- Any location
- Any time
- Through the internet

How Cloud Computing Started

◆ Amazon DDoS Attack Story

Cloud computing became popular because companies faced problems managing huge traffic.

What Happened with Amazon?

During a big sale event:

 Millions of users visited Amazon website.

At the same time:

 A DDoS attack happened.

? What is DDoS Attack?

◆ DDoS = Distributed Denial of Service

Attackers send:

- ✗ Huge unwanted traffic
- ✗ Fake requests
- ✗ Millions of packets

to the server.

🌟 Result

The server becomes overloaded:

- CPU usage becomes 100%
- RAM fills up
- Network bandwidth gets exhausted

Finally:

- ✗ Website crashes
- ✗ Services stop working

This is called:

🚨 **Server Jam / Server Down**

🧠 What Amazon Did

For the next sale:

Amazon increased servers massively.

Example:

Before	After
--------	-------

10,000 servers	100,000 servers
----------------	-----------------

Problem After Sale

After sale ended:

90,000 servers became unused.

Keeping them idle was:

 Waste of money

Amazon's Smart Idea

Amazon decided:

“Instead of keeping unused servers idle, rent them to other companies.”

This became:

 **AWS – Amazon Web Services**

Birth of Cloud Services

Now companies can:

-  Rent servers
-  Use storage
-  Create virtual machines
-  Host websites
-  Run applications

without buying physical hardware.

This is how cloud computing became popular.



CAPEX vs OPEX

◆ CAPEX (Capital Expenditure)

CAPEX means:



One-time infrastructure investment.

Example:

- Buying physical server
 - Buying racks
 - Buying switches
 - Buying storage
-



Example

Suppose you want to host your web application.

You buy:

Hardware	Cost
Server	₹2,00,000
Networking	₹50,000
Firewall	₹80,000

This is CAPEX.

◆ OPEX (Operational Expenditure)

OPEX means:



Daily operational expenses.

Examples

Expense	Example
 Electricity	Server power
 Cooling	AC for server room
 Internet	High-speed connectivity
 Staff Salary	IT engineers
 Maintenance	Hardware repair

Problem with Traditional Infrastructure

Yearly cost can become:

 ₹25–30 Lakhs+

Very expensive for small companies.

Solution = Cloud Computing

Instead of buying servers:

 Rent cloud servers

Benefits:

- Low cost
 - Easy scaling
 - No hardware maintenance
 - Fast deployment
-

Major Cloud Providers

1. AWS – Amazon Web Services

Created by:

Amazon

Market Share

AWS hosts around:

 60% of world cloud infrastructure.

2. Azure

Created by:

Microsoft

Azure is strong in:

- Enterprise environments
 - Windows servers
 - SQL databases
 - Active Directory integration
-

3. GCP – Google Cloud Platform

Created by:

Google

Popular for:

- AI
- Big Data
- Analytics
- Google services

Real Company Scenario

Suppose you need:

- Xeon Processor
- 128 GB RAM
- 8 TB storage
- GPU server

Buying physically is expensive.

Instead:

- ✓ Rent server from AWS/Azure/GCP

Now you only pay:

- 💰 Monthly rent

How Cloud Data Centers Work

Cloud companies have:

- ✓ Huge data centers
- ✓ Thousands of racks
- ✓ Multiple server types

Types of Servers Inside Cloud

Server Type	Purpose
 Web Server	Websites
 File Server	File sharing
 Database Server	Databases
 Game Server	Gaming

Server Type	Purpose
 VM Server	Virtual machines
 Mail Server	Email

Cloud Service Models

Cloud computing mainly provides:

Service Model Meaning

 SaaS	Software as a Service
 IaaS	Infrastructure as a Service
 PaaS	Platform as a Service

1. SaaS – Software as a Service

◆ Definition

SaaS provides:

- ✓ Ready-to-use software through internet.

You only:

- 👉 Open browser
- 👉 Login
- 👉 Use application

No installation needed.

◆ Cloud Provider Manages Everything

Provider manages:

- Application
 - Database
 - Runtime
 - Middleware
 - OS
 - Virtualization
 - Servers
 - Storage
 - Networking
-



User Responsibility

User only:



Uses application



SaaS Examples

Application	Type
Google Drive	Storage
Microsoft Teams Meetings	
Gmail	Email
Salesforce	CRM
ServiceNow	ITSM

Why SaaS?

Benefits:

- ✓ No installation
 - ✓ Automatic updates
 - ✓ Accessible anywhere
 - ✓ Subscription-based
 - ✓ Easy to use
-

Real-Life Example

Using Gmail:

You do NOT manage:

- servers
- operating system
- storage
- networking

Google manages everything.

You only:

 Send and receive emails.

2. IaaS – Infrastructure as a Service

◆ Definition

IaaS provides:

- ✓ Infrastructure on rent.

You get:

- Virtual machines
- Storage

- Networking
- Firewalls
- Servers

◆ Cloud Provider Manages

Managed By Cloud

Virtualization

Servers

Storage

Networking

◆ User Manages

Managed By User

Applications

Data

Runtime

Middleware

Operating System



IaaS Examples

Service	Provider
AWS EC2	Amazon
Azure VM	Microsoft
Google Compute Engine	Google

Why IaaS?

Benefits:

- ✓ Full control
- ✓ Scalable
- ✓ No hardware purchase
- ✓ Fast VM deployment
- ✓ Best for system admins

Real-Life Example

You create:

- Windows Server VM
- Install Active Directory
- Configure DNS

Cloud only gives hardware.

You manage everything else.

3. PaaS – Platform as a Service

◆ Definition

PaaS provides:

- ✓ Complete development platform.

Developers can:

- write code
- test apps
- deploy apps
- debug applications

without managing infrastructure.

◆ User Manages

Managed By User

Application

Data

◆ Cloud Manages

Managed By Cloud

Runtime

Middleware

Operating System

Virtualization

Servers

Storage

Networking

🔧 PaaS Examples

Platform	Company
Azure App Service	Microsoft
AWS Elastic Beanstalk	Amazon
Heroku	Heroku
Firebase	Google

Why PaaS?

Benefits:

- ✓ Faster development
- ✓ No server management
- ✓ Automatic updates
- ✓ Easy deployment
- ✓ Built-in scaling

Real-Life Example

A developer creates website code and uploads it to Heroku.

Heroku handles:

- server
- runtime
- OS
- scaling

Developer only focuses on coding.

SaaS vs PaaS vs IaaS Comparison

Feature	SaaS	PaaS	IaaS
Application Managed By	Cloud	User	User
OS Managed By	Cloud	Cloud	User
Server Managed By	Cloud	Cloud	Cloud
Best For	End Users	Developers	System Admins
Control Level	Low	Medium	High
Examples	Gmail	Heroku	AWS EC2

Easy Trick to Remember

Model Remember As

SaaS Ready-made software

PaaS Platform for coding

IaaS Raw infrastructure

Cloud Computing Advantages

Benefit	Explanation
 Anywhere Access	Use from any location
 Cost Saving	No hardware investment
 Scalability	Increase/decrease resources
 Backup	Data redundancy
 Fast Deployment	Create servers quickly
 Maintenance Free	Cloud handles hardware

Cloud Computing Challenges

Challenge	Explanation
 Internet Required	No internet = no access
 Security Concerns	Data stored online
 Monthly Billing	Ongoing subscription
 Downtime	Cloud outage impacts users

Cloud & Cyber Security

Cloud security is extremely important because:

- company data stored online
- applications accessible globally
- attackers target cloud infrastructure

Important Cloud Security Concepts

Concept	Purpose
IAM	Identity & Access Management
MFA	Multi-factor authentication
Encryption	Data protection
Firewall	Traffic filtering
Security Groups	Cloud firewall rules

Interview Questions

What is Cloud Computing?

Cloud computing is the delivery of computing services like servers, storage, networking, and software over the internet.

Difference Between CAPEX and OPEX?

CAPEX	OPEX
Initial investment	Operational cost
Buying server	Electricity bill
One-time cost	Recurring cost

? What is SaaS?

Ready-to-use cloud software managed completely by provider.

? What is IaaS?

Cloud provider gives infrastructure; user manages OS and applications.

? What is PaaS?

Cloud platform for developers to build and deploy applications.

🎯 Final Conclusion

This lecture covered:

- ✓ Cloud Computing Basics
- ✓ History of Cloud
- ✓ DDoS Attack Concept
- ✓ CAPEX vs OPEX
- ✓ SaaS, PaaS, IaaS

These concepts are extremely important for:

- ☁ Cloud Computing
- 🛡 Cyber Security
- 💻 System Administration
- 🌐 Networking
- 🛡 SOC Operations
- 🧑 DevOps
- 💻 IT Support



Chapter 19 – Cloud Computing



Q & A

☁️ 1. What is Cloud Computing? 🌐



Answer:

Cloud Computing means delivering IT services such as:

- Servers 🖥️
- Storage 📁
- Networking 🌐
- Applications 📦
- Databases 🗄️
- Virtual Machines 💻

through the internet instead of using local physical infrastructure.

In simple words:

👉 Instead of buying expensive hardware, companies rent resources from cloud providers like:

- ☁️ AWS
 - ☁️ Microsoft Azure
 - ☁️ Google Cloud Platform (GCP)
-



Simple Explanation:

Earlier companies had to:

- Buy servers 💰
- Maintain hardware 🔧

- Pay electricity ⚡
- Manage cooling ❄️

Now cloud providers manage everything.

Companies only:

- ✓ Pay for usage
- ✓ Access resources online
- ✓ Scale anytime

💡 Real-Life Example:

Using Google Drive:

You do NOT manage:

- Server
- Storage disks
- Network

Google handles all backend infrastructure.

You only upload/download files 📁

☁️ 2. Why Do Companies Use Cloud Computing? 🏢

✓ Answer:

Companies use cloud computing because it provides:

◆ Cost Saving 💰

No need to buy expensive servers.

◆ Scalability ⚡

Resources can increase/decrease anytime.

◆ Remote Access 🌐

Employees can work from anywhere.

◆ Backup & Disaster Recovery 🗄️

Cloud keeps redundant backups.

◆ Fast Deployment 🚀

Servers can be created within minutes.

☁️ 3. What is the Difference Between Traditional Infrastructure and Cloud?



Feature	Traditional Infrastructure 🖥️	Cloud Computing ☁️
Hardware Purchase Required	Required 🛒	Not Required ✅
Scalability	Slow ❌	Fast ✅
Maintenance	Company manages 🔧	Cloud provider manages ☁️
Cost	High upfront CAPEX 💰	Pay-as-you-go 💳
Deployment	Slow	Quick 🚀
Access	Limited	Anywhere 🌐

☁️ 4. What is AWS? 🌐

✅ Answer:

AWS (Amazon Web Services) is a cloud computing platform created by Amazon.

It provides:

- Virtual Servers 🖥️
- Storage 📁
- Networking 🌐
- Databases 🗄️
- Security 🗄️

- AI Services 🤖

through the internet.

🧠 Simple Explanation:

AWS rents infrastructure to companies.

Instead of buying physical servers:

- ✓ Companies rent virtual servers from AWS.
-

💡 Examples of AWS Services:

Service	Purpose
EC2	Virtual machines
S3	Cloud storage
RDS	Database service
IAM	Identity management

☁️ 5. What is Microsoft Azure? 🔵

✓ Answer:

Microsoft Azure is a cloud computing platform developed by Microsoft.

It is commonly used in:

- Enterprise companies 🏢
 - Windows Server environments 💻
 - Active Directory integration 🔑
 - Office 365 environments 📧
-

Real-Life Scenario:

If a company already uses:

- Windows Server
- Active Directory
- Office 365

Then Azure becomes the preferred cloud platform.

6. What is Google Cloud Platform (GCP)?

Answer:

Google Cloud Platform is Google's cloud service platform.

It is popular for:

- Big Data 
 - AI/ML 
 - Analytics 
 - Kubernetes 
-

7. Explain CAPEX vs OPEX

CAPEX (Capital Expenditure)

Answer:

CAPEX means one-time infrastructure investment.

Examples:

- Buying servers 
 - Buying firewall 
 - Buying switches 
-

💡 Example:

Company buys:

- Server = ₹2 lakh
- Firewall = ₹80k
- Storage = ₹1 lakh

This is CAPEX.

◆ OPEX (Operational Expenditure)

✅ Answer:

OPEX means daily operational expenses.

Examples:

- Electricity ⚡
 - Cooling ❄️
 - Staff salary 💻
 - Internet 🌐
 - Maintenance 🔧
-

☁️ 8. What is SaaS? 📦

✅ Answer:

SaaS means:

Software as a Service

In SaaS, everything is managed by the cloud provider. User only uses the application.

Users simply:

- ✅ Login
- ✅ Use application

without installing or managing infrastructure.

💡 Examples:

- Gmail 
- Google Drive 
- Microsoft Teams 
- Zoom 
- ServiceNow 

☁️ 9. What is IaaS?

✅ Answer:

IaaS means:

Infrastructure as a Service

Cloud provider gives:

- Virtual Machines 
- Storage 
- Networking 

User manages:

- Operating System
 - Applications
 - Data
-

Examples:

Service	Provider
AWS EC2	Amazon
Azure VM	Microsoft
Google Compute Engine	Google

Real-Life Example:

You create a Windows Server VM in AWS and install:

- Active Directory
- DNS
- DHCP

Cloud provider only gives hardware infrastructure.

10. What is PaaS?

Answer:

PaaS means:

Platform as a Service

It provides a platform for developers to:

- Write code
- Deploy apps
- Test applications

without managing servers or OS.

💡 Examples:

- Heroku
- Azure App Service
- AWS Elastic Beanstalk
- Firebase

🧠 Real-Life Example:

Developer uploads website code to Heroku.

Heroku manages:

- OS
- Runtime
- Scaling
- Server

Developer only focuses on coding 🧑💻

☁️ 11. Difference Between SaaS, PaaS, and IaaS 🔥

Feature	SaaS 📦	PaaS 🧑💻	IaaS 💻
User Control	Lowest	Medium	Highest
OS Managed By Cloud		Cloud	User
Best For	End Users	Developers	Sys Admins
Example	Gmail	Heroku	AWS EC2

💡 12. What is Virtualization? 💻

✅ **Answer:**

Virtualization means creating virtual machines using software instead of physical hardware.

One physical server can run multiple VMs.

💡 **Examples:**

- VMware
 - Hyper-V
 - VirtualBox
-

🧠 **Why Important?**

It helps:

- ✅ Save hardware cost
 - ✅ Better resource utilization
 - ✅ Easy testing/lab setup
-

💡 13. What is a Virtual Machine (VM)? 💻

✅ **Answer:**

A VM is a software-based computer that behaves like a real physical computer.

It has:

- Virtual CPU
 - Virtual RAM
 - Virtual Disk
 - Virtual Network
-

💡 Real-Life Example:

In VMware:

- Windows Server VM
- Windows 11 VM

can run on one laptop.

☁️ 14. What is DDoS Attack? 🚨

✅ Answer:

DDoS stands for:

Distributed Denial of Service

Attackers send huge fake traffic to overload servers.

🌟 Impact:

- CPU becomes 100%
 - RAM overloaded
 - Website crashes
-

🧠 Interview Explanation:

“DDoS attacks exhaust server resources using massive fake traffic.”

☁️ 15. How Does Cloud Help Against DDoS? 🛡️

✅ Answer:

Cloud platforms provide:

- Auto scaling ⚡
- Load balancing 🌐
- CDN protection 🔒

- DDoS protection services 

This helps absorb large traffic loads.

16. What is Auto Scaling?

Answer:

Auto Scaling automatically increases or decreases cloud resources based on traffic.

Example:

During online sale:

- Traffic increases 
- Cloud creates extra servers automatically 

Traffic decreases:

- Extra servers removed automatically 
-

17. What is IAM in Cloud?

Answer:

IAM stands for:

Identity and Access Management

Used to control:

- Who can login 
 - What they can access 
 - Permissions/Roles 
-

Example:

- HR user can access HR data only

- IT admin gets full access
-

☁️ **18. What is MFA?** 🔑

✅ **Answer:**

MFA means:

Multi-Factor Authentication

User must provide:

- Password 🔑
PLUS
 - OTP / Authenticator App 📱
-

🧠 **Why Important?**

Even if password is stolen:

❌ Attacker still cannot login easily.

☁️ **19. What is Cloud Storage?** 📁

✅ **Answer:**

Cloud storage means storing files on internet-based servers instead of local disk.

💡 **Examples:**

- Google Drive
 - OneDrive
 - Dropbox
 - AWS S3
-

☁️ 20. What is Load Balancer? 🌐

✅ **Answer:**

A Load Balancer distributes traffic across multiple servers.

🧠 **Purpose:**

Prevents:

- Server overload ❌
- Downtime ❌

Improves:

- Availability ✅
 - Performance ✅
-

☁️ 21. What is Disaster Recovery in Cloud? 🚨

✅ **Answer:**

Disaster Recovery means recovering systems/data after:

- Crash
- Cyber attack
- Hardware failure
- Natural disaster

Cloud helps by:

- ✅ Keeping backups
 - ✅ Replicating data
-

☁️ 22. What is Cloud Backup? 💾

✅ Answer:

Cloud backup stores copies of important data on remote cloud servers.

If system crashes:

- ✅ Data can be restored easily.
-

☁️ 23. What is Hybrid Cloud? 🌐

✅ Answer:

Hybrid cloud means using:

- Private infrastructure 🏢
AND
- Public cloud ☁️

together.

💡 Example:

Company stores sensitive data locally but hosts website on AWS.

☁️ 24. What is Public Cloud? 🌍

✅ Answer:

Public cloud is shared cloud infrastructure available to everyone.

Examples:

- AWS
 - Azure
 - GCP
-

☁️ 25. What is Private Cloud? 🏢

✅ Answer:

Private cloud is dedicated cloud infrastructure for one organization only.

Used for:

- Banking 🏦
- Government 🏛️
- High security environments 🔒

☁️ 26. What is DNS in Cloud? 🌐

✅ Answer:

DNS converts domain names into IP addresses.

Example:

google.com → IP Address

Without DNS:

❌ Browsers cannot locate websites.

☁️ 27. What is CDN? 🚀

✅ Answer:

CDN stands for:

Content Delivery Network

It stores website content in multiple locations globally.

🎯 Benefits:

- ✅ Faster website loading
- ✅ Reduced latency
- ✅ DDoS protection

🧠 28. What is Cloud Security? 🔒

✅ Answer:

Cloud security means protecting:

- Cloud servers
- Applications
- Data
- User accounts

from cyber attacks.

🔥 Important Cloud Security Controls:

- IAM 🔒
- MFA 📱
- Encryption 🔒
- Firewall 🔥
- Security Groups 🌐

🧠 29. What are Security Groups in AWS? 🔥

✅ Answer:

Security Groups act like cloud firewalls.

They control:

- ✅ Incoming traffic
- ✅ Outgoing traffic

based on rules.

☁️ **30. Explain Shared Responsibility Model** 🤝

✅ **Answer:**

In cloud:

Cloud Provider manages:

- Hardware
- Networking
- Physical security

Customer manages:

- Data
 - Passwords
 - User permissions
 - Applications
-

☁️ **31. Scenario-Based Interview Question** 🔥

❓ **A company website crashes during huge traffic. What cloud solution would you suggest?**

✅ **Professional Answer:**

I would recommend:

- 1 Auto Scaling ⚡
- 2 Load Balancer 🌐
- 3 CDN 🚀
- 4 DDoS Protection 🛡️
- 5 Monitoring tools 📊

This ensures:

- ✅ High availability
- ✅ Better performance
- ✅ Protection from overload

 32. Troubleshooting Question

 **User cannot access cloud application. What will you check?**

 **Step-by-Step Answer:**

- 1** Check internet connectivity 
- 2** Verify login credentials 
- 3** Check MFA issue 
- 4** Verify account permissions 
- 5** Check browser cache/cookies 
- 6** Check cloud service outage 
- 7** Test from another device 

 33. Job-Oriented Practical Interview Question

 **How would you deploy a Windows Server VM in AWS?**

 **Professional Answer:**

Step 1:

Login to AWS Console

Step 2:

Go to EC2 service

Step 3:

Click Launch Instance

Step 4:

Select Windows Server AMI

Step 5:

Choose VM size:

- CPU
- RAM

Step 6:

Configure:

- Storage
- Security Group
- RDP port

Step 7:

Launch VM

Step 8:

Connect using Remote Desktop Protocol (RDP)

 **34. Why is Cloud Important for IT Support?** 

 **Answer:**

Cloud knowledge is important because modern companies use:

- Office 365
- Azure
- AWS
- Google Workspace
- Cloud servers

IT Support engineers must know:

-  User management
 -  Cloud login troubleshooting
 -  MFA troubleshooting
 -  Cloud permissions
 -  VM management
-